Consolidated HIV guidelines

Service delivery

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Abbreviations

AIDS	acquired immunodeficiency syndrome
ANC	antenatal care
ART	antiretroviral therapy
DSD	differentiated service delivery
GRADE	Grading of Recommendations Assessment, Development and Evaluation
LGBTQI	lesbian, gay, bisexual, transgender, queer (or questioning), and intersex
M&E	monitoring and evaluation
mhGAP	mental Health Gap Action Programme
NCDs	noncommunicable diseases
PNC	postnatal care
SRHR	sexual and reproductive health and rights
TB	tuberculosis
U=U	undetectable = untransmissible

Definition of key terms

A public health approach	A public health approach addresses the health needs of a population or the collective health status of people rather than focusing primarily on managing individual cases. This approach aims to ensure the widest possible access to high-quality services and medicines at the population level, based on simplified and standardized approaches, and to strike a balance between implementing the best-proven standard of care and what is feasible on a large scale in resource-limited settings. In the context of HIV treatment, this includes simplified treatment regimens that work across populations, free care at the point of service, decentralized and integrated services with task-sharing, and simplified approaches to clinical monitoring
Adherence	Adherence is the extent to which a person's behaviour – such as taking medication, following a diet and/or changing lifestyle – aligns with the recommendations agreed upon with a health care provider.
Age groups	The following definitions for adults, adolescents, children and infants are used in these guidelines for the purpose of implementing recommendations for specific age groups. It is acknowledged that countries may have other definitions under national laws. An adult is a person older than 19 years of age. An adolescent is a person 10–19 years of age inclusive. A child is a person one year to less than 10 years of age. An infant is a child less than one year of age.
ART (antiretroviral therapy)	ART refers to a combination of ARV drugs used to treat HIV infection.
ARV (antiretroviral)	ARV drugs refer to the medicines used to treat or prevent HIV.
Community health worker	A community health worker is a member of the community who is trained to carry out functions related to delivering health care, health promotion and linking the community with the health system, but who does not necessarily have a formal professional or paraprofessional certificate.
Enhanced adherence support	Enhanced adherence support is a structured approach that provides additional, tailored support to individuals who are not adhering well to treatment, and which aims to identify and address barriers to adherence through counselling, education and follow-up.
Integrated service delivery	Integrated health services are services that are managed and delivered in a way that ensures people receive a continuum of health promotion, disease prevention, diagnosis, treatment, disease management, rehabilitation and palliative care services at the different levels and sites of care within the health system and according to their needs throughout the life-course. For the purposes of this guideline, the level of integration can vary from close collaboration between HIV and other services to fully integrated chronic disease management.

Key populations	Key populations are groups that have a high risk and disproportionate burden of HIV in all epidemic settings. They frequently face legal and social challenges that increase their vulnerability to HIV, including barriers to accessing HIV prevention, diagnosis, treatment and other health and social services. Key populations include men who have sex with men, people who inject drugs, people in prisons and other closed settings, sex workers and trans and gender diverse people.
Lay provider	A lay provider is any person who performs functions related to health care delivery and has been trained to deliver specific services but does not possess a formal professional or paraprofessional certificate or tertiary degree.
Nonphysician clinician	A nonphysician clinician (or health care worker) is a professional health care worker who is able to offer many of the diagnostic and clinical functions of a physician but is not trained as a physician. These types of health care workers are often known or as health officers, clinical officers, physician assistants, nurse practitioners or nurse clinicians and are an important cadre for HIV care and treatment in some countries.
Nurse	A nurse is someone who has been authorized to practise as a nurse or trained in basic nursing skills such as registered nurses, clinical nurse specialists, licensed nurses, auxiliary nurses, dental nurses and primary care nurses.
Person-centred care	Person-centred care is an approach to care that consciously adopts the perspectives of individuals, caregivers, families and communities as participants in, and beneficiaries of, trusted health systems organized around the comprehensive needs of people rather than individual diseases, and respects social preferences.
Point-of-care testing	Point-of-care testing is conducted at the site at which clinical care is being provided, with results being returned to the person being tested or caregiver on the same day as sample collection and testing in order to enable clinical decisions to be made in a timely manner.
Rapid ART initiation	Initiation of ART within seven days of HIV diagnosis.
Task sharing	Task-sharing is the rational redistribution of tasks between cadres of health care workers with longer training and other cadres with shorter training such as lay providers, while providing ongoing supervision and support.
Viral suppression	There are three key categories for HIV viral load measurements: unsuppressed (>1000 copies/mL), suppressed (detected but ≤1000 copies/mL) and undetectable (viral load not detected by test used)

Executive summary

HIV remains a major public health challenge, particularly in low- and middle-income countries, despite effective treatment and prevention tools. Many evidence-based interventions are not implemented consistently or delivered too late, contributing to ongoing transmission, advanced disease and preventable deaths. Health system constraints – including workforce shortages, weak supply chains and poor service integration – further limit impact.

These guidelines provide consolidated, evidence-based recommendations to support more people-centred, integrated and equitable HIV service delivery. The aim is to improve access, strengthen continuity of care and accelerate progress towards ending AIDS as a public health threat. They consolidate all WHO HIV service delivery recommendations issued from 2013 to 2025. New recommendations will be incorporated into this consolidated guideline and indicated accordingly.

These guidelines are intended to inform the development of relevant national and subnational health policies, clinical guidelines and programmatic guides. Therefore, the primary audience for these guidelines includes health professionals who are responsible for developing national and local health guidelines and protocols related to delivering HIV treatment and care services. The guidelines will also be of interest to hospital administrators, nursing managers, infectious and chronic disease care providers, primary care leaders, health information managers, pharmacists, professional societies, nongovernmental organizations and community-based organizations.

These guidelines were developed following WHO's standard procedures, as described in the WHO handbook for guideline development *(1)*. To support accurate interpretation and implementation, each recommendation is labelled to indicate its development status, as defined below:

- **New:** A new topic, subgroup or intervention that needed to be covered. A new evidence synthesis was conducted and the guideline development group formulated a new recommendation.
- **Updated:** A new evidence synthesis was conducted, and the topic and new evidence synthesis was reviewed by the guideline development group. The resulting recommendation reflects the latest evidence.
- **Revalidated:** An existing WHO recommendation where there is no new, impactful evidence that warrants an update to the recommendation text. The recommendation is still valid.

All external contributors, including Guideline Development Group members and the External Review Group, completed WHO declaration of interests forms, and are listed in Annex.

Table 1: HIV service delivery recommendations

Date*	Recommendations	Status
Linkage and ART initiation		
2016	Following an HIV diagnosis, a package of support interventions should be offered to ensure timely linkage to care for all people living with HIV *(Strong recommendation, moderate-certainty evidence)*.	Revalidated
2021	Health systems should invest in people-centred practices and communication, including ongoing training, mentoring, supportive supervision and monitoring of health workers, to improve the relationships between clients and health-care providers *(Good practice statement)*.	Revalidated
2019	ART initiation should follow the overarching principles of providing people-centred care. People-centred care should be focused and organized around the health needs, preferences and expectations of people and communities, upholding individual dignity and respect, especially for vulnerable populations *(Good practice statement)*.	Revalidated
2021	Antiretroviral therapy (ART) initiation may be offered outside the health facility *(Conditional recommendation, low- to moderate-certainty evidence*).	Revalidated
2017	Rapid ART initiation should be offered to all people living with HIV following a confirmed HIV diagnosis and clinical assessment *(Strong recommendation: high-certainty evidence for adults, low-certainty evidence for children)*	Revalidated
2021	The offer of same-day ART initiation should include approaches to improve uptake, treatment adherence and retention such as tailored client education, counselling and support *(Good practice statement)*.	Revalidated
Clinical visits and medication pick-up		
2021	People established on ART should be offered clinical visits every 3–6 months, preferably every six months if feasible *(Strong recommendation, moderate-certainty evidence*).	Revalidated
2021	People established on ART should be offered refills of ART lasting 3–6 months, preferably six months if feasible *(Strong recommendation, moderate-certainty evidence)*.	Revalidated
Adherence and engagement in care		
2025	Adherence support interventions should be provided to people on ART *(Strong recommendation, moderate-certainty evidence)*.	Updated
2021	Viral load for treatment monitoring should be complemented with non-judgemental, tailored approaches to assessing adherence *(Good practice statement)*.	Revalidated
2021	HIV programmes should implement interventions to trace people who have disengaged from care and provide support for re-engagement *(Strong recommendation, low-certainty evidence)*.	Revalidated
Services for adolescents and young people		
2021	Psychosocial interventions should be provided to all adolescents and young adults living with HIV *(Strong recommendation, moderate-certainty evidence)*.	Revalidated
2013	Adolescent-friendly services should be implemented in HIV services to ensure engagement and improved outcomes *(Strong recommendation, moderate-certainty evidence)*.	Revalidated

Table 1: HIV service delivery recommendations (cont'd)

Date*	Recommendations	Status
Services for adolescents and young people		
2013	Adolescents should be counselled about the potential benefits and risks of disclosing their HIV status to others and empowered and supported to determine whether, when, how and to whom to disclose *(Conditional recommendation, very-low -certainty evidence)*.	Revalidated
Task sharing		
2016	Trained non-physician clinicians, midwives and nurses can initiate first-line ART *(Strong recommendation, moderate-certainty evidence)*.	Revalidated
2016	Trained non-physician clinicians, midwives and nurses can maintain ART *(Strong recommendation, moderate-certainty evidence)*.	Revalidated
2016	Trained and supervised community health workers can dispense ART between regular clinical visits *(Strong recommendation, moderate-certainty evidence)*.	Revalidated
2016	Trained and supervised lay health-care providers can distribute ART *(Strong recommendation, low-certainty evidence)*.	Revalidated
2021	Task sharing of specimen collection and point-of-care testing with non-laboratory personnel should be implemented when professional staffing capacity is limited *(Strong recommendation, moderate-certainty evidence)*.	Revalidated
Service Delivery models and integration		
2013	Decentralization of ART care should be considered as a way to increase access and improve retention in care *(Strong recommendation, moderate-to-low-certainty evidence)*.	Revalidated
2013	In generalized epidemic settings, ART should be initiated and maintained in pregnant and postpartum women and in infants at maternal and child health-care settings, with linkage and referral to ongoing HIV care and ART, if appropriate *(Strong recommendation, low-certainty evidence)*.	Revalidated
2021	Sexual and reproductive health services, including contraception, may be integrated with HIV services *(Conditional recommendation, very-low-certainty evidence)*.	Revalidated
2013	In settings with a high burden of HIV and tuberculosis (TB), ART should be initiated in TB treatment settings, with linkage to ongoing HIV care and ART *(Strong recommendation, very-low-certainty evidence)*.	Revalidated
2013	In settings with a high burden of HIV and TB, TB treatment may be provided for people living with HIV in HIV care settings if they have been diagnosed with TB *(Strong recommendation, very-low-certainty evidence)*.	Revalidated
2025	Diabetes and hypertension care should be integrated with HIV services *(Strong recommendation, moderate-to-very-low-certainty evidence)*.	Updated
2025	Mental health care for depression, anxiety and alcohol use disorders should be integrated with HIV services *(Strong recommendation, moderate-to-very-low-certainty evidence)*.	New

* Date the recommendation/good practice statement was formulated

1. Introduction

HIV remains a major public health challenge and a leading cause of illness and death in many low- and middle-income countries, despite the wide availability of effective antiretroviral therapy and proven prevention tools.

In many settings, evidence-informed interventions for HIV prevention, testing, treatment and long-term care are not fully implemented, delivered inconsistently or reach people too late, contributing to ongoing transmission, advanced disease and avoidable mortality. Persistent health system constraints – including workforce shortages, weak supply chains and limited integration with related services such as those for noncommunicable diseases and mental health care – further hinder effective service delivery.

These HIV service delivery guidelines provide updated and consolidated recommendations to support countries, programme managers, health-care workers, communities and other stakeholders to organize services that are people-centred, integrated, equitable and grounded in the best available evidence. The guidelines aim to improve access to and continuity of high-quality HIV services, reduce inequities in outcomes and accelerate progress towards ending AIDS as a public health threat.

2. Linkage and ART initiation

2.1 Linkage from HIV testing to enrolment in care

Recommendation

Following an HIV diagnosis, a package of support interventions should be offered to ensure timely linkage to care for all people living with HIV.
(Strong recommendation, moderate-certainty evidence)

The following interventions have demonstrated benefit in improving linkage to care following an HIV diagnosis:

- streamlined interventions to reduce time between diagnosis and engagement in care, including (i) enhanced linkage with case management; (ii) support for HIV disclosure; (iii) tracing; (iv) training staff to provide multiple services; and (v) streamlined services *(moderate-certainty evidence)*;
- peer support[a] and navigation approaches for linkage *(moderate-certainty evidence)*; and
- quality improvement approaches using data to improve linkage *(low-certainty evidence)*.

[a] Includes peer counselling.

Revalidated

Good practice statements

ART initiation should follow the overarching principles of providing people-centred care. People-centred care is care that is focused and organized around the health needs, preferences and expectations of people and communities, upholding individual dignity and respect, especially for vulnerable populations. It should promote the engagement and support of people and families to play an active role in their own care through informed decision-making.

All people newly diagnosed with HIV should be retested to verify their HIV status before starting ART, using the same testing strategy and algorithm as the initial test. To minimize the risk of misdiagnosis, this approach should be maintained in settings in which rapid ART initiation is being implemented.

Revalidated

Background

Linkage to HIV prevention, treatment and care after testing remains a challenge, with disproportionate gaps for key populations, men, young people and previously diagnosed people living with HIV (not receiving ART, disengaged or lost to care).

HIV testing service providers are primarily responsible for linkage to care. People testing HIV positive should be assessed for advanced HIV disease using CD4 cell count testing and offered ART, and those testing negative should be linked to prevention services.

Rationale

Evidence-informed interventions to support linkage can be classified into three approaches *(2)*:

- streamlined services via enhanced linkage with case management, HIV disclosure support, client tracing for non-engagers and staff training for multiple services;
- peer support and navigation interventions, including home visits, system navigation and enhanced counselling; and
- quality improvement interventions enhance linkage to care *(3)*.

Implementation considerations

- Interventions may vary based on the context, including health-care delivery systems, geography and target population.
- A combination of interventions may be needed to improve linkage for specific underserved groups, especially key populations and men.
- Barriers to linkage to HIV care and ART initiation should be assessed and addressed, especially for populations facing multiple structural and individual barriers.
- Linkage for children can be improved using point-of-care early infant diagnosis, SMS/GSM/GPRS printers to accelerate access to laboratory results and family-centred models delivering care to mother, baby, and partner at the same point.
- Health facilities should have capacity to provide services for key populations. Community- and peer-based approaches can be especially effective in restrictive legal settings.

WHO guidelines on HIV testing services *(4)* provide information on post-test services and linkage to care.

2.2 People-centred care

Good practice statement

Health systems should invest in people-centred practices and communication, including ongoing training, mentoring, supportive supervision and monitoring of health workers, to improve the relationships between clients and health-care providers.

HIV programmes should:

- **provide people-centred care that is focused and organized around the health needs, preferences and expectations of people and communities, upholding individual dignity and respect, especially for underserved populations, and engage and support people and families to play an active role in their own care by informed decision-making;**
- **offer safe, acceptable and appropriate clinical and non-clinical services in a timely fashion, aiming to reduce morbidity and mortality associated with HIV infection and to improve health outcomes and quality of life in general; and**
- **promote the efficient and effective use of resources.**

Revalidated

Background

People-centred health services are an approach to care that consciously adopts the perspectives of individuals, families and communities and sees them as participants and beneficiaries of trusted health systems that respond to their needs and preferences in humane and holistic ways. This approach acknowledges the experiences and perspectives of health-care providers that may enable or prevent the delivery of high-quality people-centred care.

Rationale

- A systematic review *(5)* identified several interventions to enhance people-centred care for people living with HIV. Overall, studies reported beneficial effects of these approaches across the HIV cascade outcomes, including improved ART uptake, adherence and viral load suppression.
- Studies have shown that people are willing to travel longer distances to be seen by a health-care provider with a respectful and caring attitude, and negative health-care worker attitudes contribute to loss to care and poor programme outcomes *(6)*.

Interventions found to improve relationships between clients and health-care providers can be classified into the following approaches:

- providing friendly and welcoming services: training providers to make general HIV services more welcoming, including for clients who have disengaged from care;
- conducting sensitization training for clinical and non-clinical health-care providers to improve care for key populations both at the primary care and community levels; such training should include issues related to stigma and discrimination;
- offering individualized adherence counselling and person-centred communication such as shared decision-making and planning for ART initiation and adherence;
- facilitating client education in empowerment and communication skills; and
- providing feedback to health-care workers on client concerns and evaluation of service quality such as community score cards and client feedback mechanisms combined with quality improvement exercises.

Implementation considerations

- Build capacity for ongoing training, mentoring and supportive supervision on people-centred communication, stigma reduction and welcoming services.
- Integrate sensitization training for all staff (clinical and non-clinical) on respectful attitudes, stigma and discrimination, re-engagement of clients who have disengaged, and support for women and other vulnerable groups experiencing violence or sexual health concerns.
- Establish routine client feedback mechanisms (such as satisfaction questionnaires, community score cards and suggestion boxes) and use the results in facility quality-improvement meetings.
- Involve clients and community representatives in designing and reviewing services.
- Review workload, staffing and client flow to allow time for person-centred interactions.

2.3 Initiating ART outside the health facility

Recommendation

ART initiation may be offered outside the health facility.

(Conditional recommendation, low- to moderate-certainty evidence)

Revalidated

Background

Community-based HIV testing approaches are a key component of any HIV testing strategy. However, losses to care between community HIV testing and referral to a health facility for ART initiation can be substantial.

Rationale

- A systematic review *(7)* reported that offering ART initiation outside the health facility increased in the proportion of people starting ART, increased retention in care at 6–12 months following ART initiation and increased suppression of viral loads.
- Another systematic review *(8)* found that rapid initiation of ART is associated with several health benefits, including reduced mortality and morbidity and reduced onward transmission.
- Implementation could increase treatment access for individuals facing structural barriers such as criminalization, stigma and discrimination when accessing health-care services to initiate treatment.

Implementation considerations

- The recommendation is additional to the routine offer of ART initiation at the health facility.
- Confirmatory HIV testing, baseline assessments and support, including ART literacy and clinical and psychosocial assessment that includes CD4 cell count and linkage to the advanced HIV disease package, should be provided either as part of the intervention or on referral to the health facility.
- Systems for ensuring future linkage to a facility providing ART services must be supported.

2.4 Rapid initiation of ART, including same-day start

Recommendations

Rapid ART initiation[a] should be offered to all people living with HIV following a confirmed HIV diagnosis and clinical assessment.

(Strong recommendation: high-certainty evidence for adults and adolescents, low-certainty evidence for children)

ART initiation should be offered on the same day to people who are ready to start.

(Strong recommendation: high-certainty evidence for adults and adolescents, low-certainty evidence for children)

a Rapid initiation is defined as within seven days from the day of HIV diagnosis.

Revalidated

Good practice statement

The offer of same-day ART initiation should include approaches to improve uptake, treatment adherence and retention such as tailored client education, counselling and support. Revalidated

Background

All people living with HIV can benefit from ART. Delays in starting treatment can result in loss to care, progression to illness and onward transmission.

Rationale

- A systematic review *(8)* showed that rapidly initiating ART, including on the same day as diagnosis, leads to an increased likelihood of starting treatment, improved suppression of viral loads and retention in care and may lead to reduced mortality.
- Clients have highlighted the importance of good counselling and non-judgemental, respectful personnel *(9)*.
- Rapid initiation of ART is especially important for people with very low CD4 cell counts, who have a high risk of death, unless there is a clinical contraindication.

A range of evidence-informed approaches, both pre- and post-ART initiation, targeting clients, health-care providers and the health system, can be implemented to support same-day ART initiation (Table 2).

Table 2. Evidence-informed approaches to supporting same-day ART initiation at the level of the client, provider and health system

	Strategies targeting clients	Strategies targeting health-care providers	Strategies targeting the health system
Before ART Initiation	Reduce administrative requirements to initiate ART	Ensure provider training on rapid ART initiation	Reduce the number of pre-ART sessions
	Reduce pre-ART psychosocial requirements	Ensure provider training on counselling	Provide the first ART counselling session on the day of HIV testing
	Improve pre-ART counselling content and delivery (consider remote and community treatment literacy)	Provide provider supervision, coaching and mentorship	Increase the quality of pre-ART sessions
	Promote shared decision-making	Provide provider performance feedback	Expedite the scheduling of appointments to initiate ART
	Reduce pharmacy waiting time	Provide standard operating procedures and guidance documents	Provide ART starter pack immediately

Table 2. Evidence-informed approaches to supporting same-day ART initiation at the level of the client, provider and health system (cont'd)

	Strategies targeting clients	Strategies targeting health-care providers	Strategies targeting the health system
Before ART Initiation	Support navigation during ART initiation visit	Provide decision support tool (checklist or algorithm)	Provide point-of-care CD4, TB and cryptococcal antigen testing as indicated and diagnosis
After ART initiation	Send appointment reminders		
	Provide short-term ongoing navigation and support		
	Provide intensified post-ART counselling		
	Provide increased duration post-ART initiation clinical visit		
	Provide an incentive to attend post-ART initiation visits		

Implementation considerations

- Treatment literacy and psychosocial readiness for initiation should be assessed. However, initiation should not be delayed to complete counselling content. Counselling should be scheduled over the initial weeks of treatment. Consider options to provide this initial counselling and assessment of treatment literacy remotely or in the community.
- Confirmatory HIV testing should be carried out. CD4 cell count testing and clinical assessment should be performed to determine whether the client has advanced HIV disease and to prevent opportunistic infections.
- Among people with TB symptoms, evidence supports starting ART while investigating for TB *(10)*. ART initiation should be delayed among people with meningitis and cryptococcal meningitis because of the increased risk of severe adverse events and immune reconstitution inflammatory syndrome. Cryptococcal meningitis is diagnosed with serum cryptococcal antigen screening and confirmed by cerebrospinal fluid analysis.
- People should not be coerced to start immediately and should be supported in making an informed choice regarding when to initiate ART.

3. Clinical visits and medication pick-up

Recommendations

People established on ART should be offered clinical visits every 3–6 months, preferably every six months if feasible.

(Strong recommendation, moderate-certainty evidence)

People established on ART should be offered refills of ART lasting 3–6 months, preferably six months if feasible.

(Strong recommendation, moderate- to low-certainty evidence)

Revalidated

Background

Frequent, clinician-led clinic visits are unnecessary for most people who are established on ART and place an avoidable burden on the health-care system. They can result in avoidable costs for clients, increased workload for providers and preventable losses to care. The duration of medication prescribing and dispensing for chronic care should be delinked from the frequency of clinical assessment.

Individuals who are established on ART are eligible for less frequent clinic visits and medication refills. The following criteria are used to determine if an individual is established on ART:

- receiving ART for at least six months;
- no current illness (excluding well-controlled chronic conditions);
- demonstrated understanding of lifelong adherence (has received adherence counselling); and
- evidence of treatment success: suppressed viral load within the past 6 months, or CD4 >200 cells/mm^3 (or >350 for children 3–5 years old) or weight gain with no symptoms or concurrent infections.

Differentiated HIV treatment delivery – providing less frequent clinical visits and longer duration of ART refills for clients established on treatment through a range of ART refill models (such as community-based drug pick-up and fast-track pick-up at facilities) – reduces client travel time and costs, shortens waiting times, is acceptable to clients and lowers health system expenses, making it a cost-saving approach *(11)*.

Rationale

- A systematic review *(12)* found that extending clinical visits from three to six months showed similar outcomes in retention in care and viral load suppression.
- Studies comparing different ART dispensing intervals also found that longer refills (up to six months) had similar retention and viral load outcomes to shorter refills and were largely comparable across studies *(12)*.

- Research from six African countries showed that children and adolescents who transitioned to multimonth prescribing maintained favourable outcomes, including survival, adherence, retention and viral suppression *(13)*.

Implementation considerations

- Reduced dispensing frequency requires reliable drug supply and adequate storage capacity for clients, including for community ART delivery.
- Multimonth prescribing can be implemented even when multimonth dispensing is not feasible due to supply limitations.
- Health-care services for chronic conditions, contraceptive care and TB preventive therapy should be integrated and schedules aligned to minimize clinical visits.
- Extended refills are recommended only for children older than two years, since younger children require more frequent visits for dose adjustments and routine services (such as immunization).
- Some higher-risk groups (such as adolescents and pregnant and postpartum women) may require adapted clinical and refill schedules or additional community support to address access challenges related to schooling and adherence challenges and to provide specific medical interventions.

4. Adherence and engagement in care

4.1 Adherence support

Recommendation

Adherence support interventions should be provided to people on ART.

(Strong recommendation, moderate-certainty evidence)

The following interventions have demonstrated benefit:

- counselling *(moderate-certainty evidence)*
- reminders *(moderate-certainty evidence)*
- peer and other support *(moderate-certainty evidence)*
- education. *(low-certainty evidence)*

Updated

Good practice statement

Viral load for treatment monitoring should be complemented with non-judgemental, tailored approaches to assessing adherence.

Revalidated

Background

Suboptimal adherence to medication is a global challenge, leading to disease progression, poor health outcomes and increased health-care costs. In HIV, suboptimal adherence also has public health risks, including an increased risk of transmission because of unsuppressed viral loads and the development of drug resistance, making future treatment more complicated and costly. People receiving ART face multiple barriers to sustained lifelong adherence, categorized as individual, interpersonal, community-based and structural barriers *(14)*.

Rationale

- A network meta-analysis found that the following interventions improve adherence and viral suppression: reminders, supporter-based strategies, education, counselling and complex packages *(15)*. Reminders were particularly effective for adolescents, and supporters and counselling helped people with adherence challenges and some mental health or substance use problems, although the evidence was limited.
- The feasibility and cost–effectiveness of ART adherence support varies by intervention type, intensity and context.

Implementation considerations

- Counselling and education have been widely integrated into national programmes, although barriers include limited training, compensation for lay workers and risks of unintended HIV disclosure. Peer and community support groups are effective and should be supported.
- To maximize impact, adherence support strategies should be directed towards individuals with the greatest risk of suboptimal adherence.
- Consideration should be given to combining or sequencing adherence interventions in terms of duration and intensity to suit the needs and preferences of the individual. Adherence should be reassessed during clinical appointments.
- The initial months following diagnosis and treatment initiation are especially critical, since disengagement from care is more likely during this period; early and targeted support during this period can improve long-term treatment outcomes.
- Preferred locations for adherence support (facility, community and telehealth) may vary by context and client preference; in some settings, clinic-based support is favoured because of lower stigma and easier access.
- Adherence support systems can be leveraged across comorbid conditions, including diabetes, hypertension and mental health, to improve the efficiency and continuity of care.
- Reminders and supporter-based strategies must be designed to avoid inadvertent HIV disclosure by using opt-in communication, neutral wording and clear consent.
- Self-care and self-management should be promoted using simplified tools, education and empowerment strategies, especially in settings with limited health system capacity.
- Explaining the goal of U=U (undetectable = untransmissible) is key to supporting ART adherence.

Simple, low-cost approaches for measuring adherence

- Pharmacy refill records provide information on when people pick up their antiretroviral drugs.
- Pill counts may help to assess adherence but may not be feasible in routine care settings. Pill count has been found to perform better when combined with self-reported adherence.
- A key value of all these approaches is that they encourage discussions about adherence with people receiving ART.

4.2 Tracing and re-engagement in care

Recommendation

HIV programmes should implement interventions to trace people who have disengaged from care and provide support for re-engagement.

(Strong recommendation, low-certainty evidence)

Revalidated

Background

Disengagement from care remains common, affecting all age groups. Many disengaged individuals are willing to return to care.

Rationale

- A systematic review *(16)* of activities to trace individuals and support re-engagement found that most people traced re-engaged in care.
- Tracing soon after a missed visit was more effective, using remote (phone, text, mail and email), in-person or combined approaches.
- Tracing activities carry the risk of inadvertently disclosing HIV status. Although this risk is small and is outweighed by the benefits of re-engaging individuals into care and onto life-saving ART, safeguards include prior consent to tracing, client preferences and risk assessment for key populations.

Implementation considerations

- A non-judgemental approach is essential to supporting people to return to care.
- Health-care providers should assess, during routine clinical encounters, factors associated with increased risk of disengagement from HIV care.
- The criteria for tracing and recall should consider those who are seven or more calendar days late for a scheduled appointment.
- Priority for tracing should focus on those who recently started treatment, who have abnormal test results, who have a history of advanced HIV disease and who are not yet receiving treatment. Other priority groups include pregnant and breastfeeding women, children and adolescents.
- Clients should be provided with the opportunity to consent to tracing when ART follow-up is discussed during counselling and at ART initiation.
- Tracing is usually carried out by health facility staff, sometimes with social or community health workers, and requires training and resources, including establishing systems to trace and support re-engagement.
- Tracing may be carried out through text messages, phone calls or visits in the community.
- The frequency of tracing interventions and when tracing will stop should be clearly defined.
- Programme- or facility-level confidential client contact details should be kept up to date to ensure successful tracing if required.
- Direct re-entry into differentiated service delivery models for clients established on treatment should be considered to facilitate retention and adherence for those who return to care within three months of their scheduled appointment.
- Understanding client-reported reasons for disengagement is essential for identifying both general and local causes of retention failures.
- Adequately assessing the risks of tracing activities for vulnerable and key populations is critical.

5. Services for adolescents and young people

Recommendations

Psychosocial interventions should be provided to all adolescents and young adults living with HIV.

(Strong recommendation, moderate-certainty evidence)

Adolescent-friendly services should be implemented in HIV services to ensure engagement and improved outcomes.

(Strong recommendation, low-certainty evidence)

Adolescents should be counselled about the potential benefits and risks of disclosing their HIV status to others and empowered and supported to determine whether, when, how and to whom to disclose.

(Conditional recommendation, very-low-certainty evidence)

Revalidated

Background

Adolescents and young adults living with HIV face distinct challenges as they assume responsibility for their own care amid stigma and disclosure concerns. Adolescence is a period of identity exploration and vulnerability, and mental health and social difficulties compound for adolescents living with HIV. Compared with adults 25 years and older, they are generally underserved by current HIV services, with poorer ART access and coverage, worse viral suppression and higher loss to follow-up before and after ART initiation.

Rationale

- A systematic review found low-certainty evidence that adolescent-friendly health services, compared with standard care, produced improvements in: health outcomes (such as lower pregnancy rates); health service use; uptake of services (notably HIV testing); knowledge about HIV, sexually transmitted infections, pregnancy prevention and sexual health; attitudes towards sex and HIV testing; sexual risk-reduction behaviour (such as condom use); self-efficacy; improved adherence and viral suppression; and acceptability of services to adolescents *(17)*.
- Another systematic review *(18)* provided evidence that psychosocial interventions improved ART knowledge, linkage to care, adherence to ART, retention in care, viral load, sexual and reproductive health behaviour and knowledge and improved transitioning to adult services.
- Psychosocial interventions for adolescents and young adults living with HIV are feasible to implement and well accepted. No undesirable effects were identified.

The following psychosocial interventions were found to provide benefit:

- interventions that harnessed motivational interviewing: a collaborative, client-centred counselling style focused on increasing motivational readiness for behavioural change *(18–22)*.
- interventions that involved adolescents and their caregivers can strengthen communication, problem-solving and negotiation skills for both adolescents and caregivers *(23)*;
- interventions based around peer support and social networks *(24–27)*; and
- digital approaches to introduce new information and deliver behaviour change skills *(27–31)*.

Implementation considerations

The global standards for quality health-care services for adolescents *(32)* provide an approach to implementing health-care services for adolescents. This publication includes nine standards covering the following areas: adolescent-centred care and empowerment; developmentally responsive care; inclusive, confidential, respectful and safe care; family and community engagement; competent human resources; comprehensive health benefit package of care; data-informed and youth-engaged practice; a welcoming physical environment; and accessible service delivery platforms.

The following implementation considerations should be considered:

- a package of services that is both acceptable and feasible and differentiated according to the needs and experiences of subpopulations;
- linkage and referral pathways to ensure a comprehensive continuum of care, especially for the transition from paediatric or adolescent services to adult HIV services;
- interventions can be delivered through a range of delivery modalities, including clinic visits, home visits, support groups (including peer support and groups that link psychosocial support with ART delivery such as teen clubs), social media and telephone contact;
- the location of the intervention also influenced feasibility, with interventions delivered digitally or at home considered more feasible because of convenience and flexibility;
- comprehensive training of existing or new personnel and integrating interventions into existing health-care settings;
- peer support is especially important for managing the health of adolescents and young adults living with HIV and interventions that focus on strengthening support from trusted family members and health-care workers are also desired;
- differences in exposure to risks and protective factors depending on age, developmental stage, sex, health status, whether they belong to a key population and context;
- maintaining the highest ethical standards, including voluntary participation, confidentiality, privacy and the best interests of each adolescent and young person, with failure to participate not affecting access to ART or other services;
- community support and the involvement of parents, guardians and other community members in programmes may provide important support for programmes and promote their success.

6. Task sharing

Recommendations

The following recommendations apply to all adults, adolescents and children living with HIV.

- **Trained non-physician clinicians, midwives and nurses can initiate first-line ART.**
(Strong recommendation, moderate-certainty evidence)
- **Trained non-physician clinicians, midwives and nurses can maintain ART.**
(Strong recommendation, moderate-certainty evidence)
- **Trained and supervised community health workers can dispense ART between regular clinical visits.**
(Strong recommendation, moderate-certainty evidence)
- **Trained and supervised lay health-care providers can distribute ART.**
(Strong recommendation, low-certainty evidence)

Revalidated

Background

Many settings have insufficient numbers of health-care providers. Task sharing involves redistributing tasks within health workforce teams. Most people living with HIV on ART are well and can receive ART with minimal clinical support.

Rationale

- A systematic review *(33)* found that ART started or managed by non-physician clinicians results in similar clinical outcomes and lower loss to follow-up compared with physician-led care.

Implementation considerations

- Initial and ongoing training, mentoring, supportive supervision and effective administrative planning are critical.
- Adapted clinical guidance for specific cadres with clear pathways for treatment and referral are required to support high-quality care.
- Adopting task-sharing strategies often requires revising regulatory frameworks and national policies to enable new cadres of health-care workers to perform additional tasks.
- WHO recommends that all health-care workers, including community health workers, be appropriately trained, remunerated and supervised *(34)*.
- National regulatory bodies, professional associations and other stakeholders need to be involved in defining the scope of practice, roles and responsibilities of health-care workers.

Recommendation

Task sharing of specimen collection and point-of-care testing with non-laboratory personnel should be implemented when professional staffing capacity is limited.

(Strong recommendation, moderate-certainty evidence)

Revalidated

Background

The lack of skilled laboratory professionals in health-care facilities may require sharing the tasks of point-of-care diagnostic testing and sample collection with lower cadres of health-care workers. Task sharing may also increase access to testing for key populations, who may have difficulty in accessing traditional health-care services regardless of professional staffing capacity.

Rationale

- A systematic review found moderate-certainty evidence that a wide range of non-laboratory health workers, especially nurses, can perform multiple types of point-of-care tests with good diagnostic accuracy. The performance of point-of-care CD4 testing performed by non-laboratory health workers does not differ significantly by cadre or versus laboratory testing *(35)*.
- Another systematic review *(36)* found that point-of-care infant testing technologies had very high diagnostic accuracy, with similar failure rates and result return when tests were performed by nurses versus laboratory-trained personnel. Time from sample collection to ART initiation among infants found to be HIV-positive was significantly lower with point-of-care testing. Additional evidence supports task sharing across a range of diagnostic testing, including cryptococcal antigen lateral flow assays, syphilis testing and routine blood tests.
- Task sharing is feasible and acceptable. In a study conducted across eight African countries, health-care workers and programme managers reported that point-of-care infant testing was easy to use and highly acceptable, especially because rapid results enabled infants living with HIV to start treatment sooner *(37)*.

Implementation considerations

- Decentralization and task sharing with non-laboratory personnel should occur in the context of a deliberate expansion of human resource capacity and, in some settings, updated legal and regulatory frameworks.
- Decentralized specimen collection and point-of-care testing require strong central support for training, mentorship, equipment maintenance, quality assurance and accurate data capture.
- Access to high-quality diagnostics across HIV and other molecular needs should be expanded using integrated laboratory networks that combine centralized and point-of-care technologies.

7. Service Delivery models and integration

7.1 Decentralization

Recommendation

Decentralization of ART care should be considered as a way to increase access and improve retention in care.

The following approaches have demonstrated effectiveness in improving access and retention:

initiating ART in hospitals, with maintenance of ART in peripheral health facilities

(Strong recommendation, low-certainty evidence)

initiating and maintaining ART in peripheral health facilities

(Strong recommendation, low-certainty evidence)

initiating ART at peripheral health facilities, with maintenance at the community level.[a]

(Strong recommendation, moderate-certainty evidence)

See also the recommendation that ART initiation may be offered outside the health facility

[a] The community level includes external outreach sites, health posts, home-based services or community-based organizations. The frequency of clinic visits depends on health status.

Revalidated

Background

Decentralizing HIV care and treatment can decongest facilities, bring services closer to people's homes, strengthen community engagement and improve access, care-seeking and retention in care.

Rationale

- A systematic review *(38)* provides evidence that decentralizing ART care, either from hospitals to health centres or from health centres to community-based care, reduces attrition without compromising clinical outcomes. The benefits of decentralization may differ by setting.

Implementation considerations

- The optimal decentralization model depends on local disease burden and health system capacity. Partial decentralization models include specialist-led shared care, teleconsultation, nurse-led follow-up and community refill sites.

- Programme managers should consider client preferences, expected volumes and whether decentralization brings services closer to people; community engagement should be supported to ensure person-centred decisions.
- Standards of care, roles and responsibilities should be clearly defined at each level, including the private sector.
- Decentralization must be supported by strong linkage and referral systems and adequate laboratory services, monitoring and evaluation and supply management.
- Decentralized services should be adapted for key populations, pregnant and postpartum women and HIV-exposed children and children living with HIV.

7.2 Delivering ART in maternal and child health-care settings

Recommendation

In generalized epidemic settings, ART should be initiated and maintained among pregnant and postpartum women and among infants at maternal and child health-care settings, with linkage and referral to ongoing HIV care and ART, if appropriate.

(Strong recommendation, low-certainty evidence)

Revalidated

Background

Maternal and child health settings provide a key opportunity to provide access to HIV treatment and care. Integrating HIV testing, timely ART initiation, infant diagnosis and intensified viral load monitoring into maternal and child health services and ensuring effective referral to long-term HIV care provide a critical, person-centred opportunity to close gaps in care and reduce vertical transmission.

Rationale

- A systematic review *(39)* found that integrating HIV and maternal and child health services improves ART uptake and adherence in pregnancy, with maternal and infant outcomes comparable to or better than non-integrated care.
- Health-care providers report that integrated services increased efficiency, reduced client waiting times and stigma and strengthened provider–client relationships, supporting better adherence to treatment and care *(40)*.

Implementation considerations

- In many contexts, most women living with HIV who become pregnant are already receiving ART. These women should be offered the choice of continuing to receive their HIV care in their differentiated service delivery model while attending antenatal care, postnatal care and receiving additional viral load monitoring.
- Clear plans are needed for when and how to transition and link families to HIV services and, where relevant, back into maternal and child health care.

- Decisions should preserve women's choice of ART care location and consider service capacity, quality, accessibility and local HIV burden, without compromising efficient, person-centred antenatal care.
- Differentiated service delivery models (such as adherence clubs) may be adapted for postpartum care of the mother and the exposed infant.

7.3 Integrating sexual and reproductive health services, including contraception, with HIV services

Recommendation

Sexual and reproductive health services, including contraception, may be integrated with HIV services.

(Conditional recommendation, very-low-certainty evidence)

Revalidated

Background

Many women of reproductive age want contraceptive care, but lack access to effective contraception. Sex workers often have greater unmet contraceptive needs and tend to rely mainly on condoms rather than combining methods for better protection. Access to modern sexual and reproductive health services has improved globally, but many people still have unmet contraceptive care needs. Women living with HIV frequently face stigma and discrimination, making supportive environments crucial for better health outcomes. Preventing unintended pregnancies is one strategy for preventing vertical HIV transmission.

Rationale

- One systematic review *(41)* found that integration was associated with higher prevalence and knowledge of modern methods of contraception. A second review *(42)* found that integrating HIV testing into family planning services is feasible and has potential for positive joint outcomes.

Implementation considerations

- Integration requires suppport to overcome barriers to service uptake, use and engagement. Barriers exist at the individual, interpersonal, community and societal levels, including social exclusion, marginalization, criminalization, stigma, gender-based violence and gender inequality.
- Training on human sexuality promotes understanding of sexually diverse communities, especially LGBTQI people and adolescents and young adults seeking accurate sexual and reproductive health and rights information and services.
- Legal and policy barriers to sexual and reproductive health and rights services should be addressed, especially for adolescents.
- WHO strongly recommends integrating care for intimate partner violence and sexual assault into existing health services *(43)*.

7.4 Delivering ART in TB treatment settings and TB treatment in HIV care settings

Recommendations

In settings with a high burden of HIV and TB, ART should be initiated in TB treatment settings, with linkage to ongoing HIV care and ART.

(Strong recommendation, very-low-certainty evidence)

In settings with a high burden of HIV and TB, TB treatment may be provided for people living with HIV in HIV care settings if they have also been diagnosed with TB.

(Strong recommendation, very-low-certainty evidence)

Revalidated

Background

TB is a leading cause of illness and death among people living with HIV. Access to treatment for both diseases needs to be ensured as soon as indicated.

Rationale

- A systematic review *(44)* found evidence of increased uptake and timeliness of ART initiation when ART was delivered in TB treatment settings and decreased mortality when TB treatment was delivered in HIV care settings. TB treatment success rates and ART uptake were comparable across studies.

7.5 Integrating HIV, diabetes and hypertension care

Recommendation

Diabetes and hypertension care should be integrated with HIV services.

(Strong recommendation, moderate-certainty evidence for blood pressure control and very-low-certainty evidence for diabetes control)

Updated

Background

The number of older adults living with HIV is growing as the population of people living with HIV ages, and conditions such as hypertension and diabetes are common in this population, especially in sub-Saharan Africa. High blood pressure is a major risk factor for death. The number of people living with hypertension has risen sharply in recent decades; diabetes also affects hundreds of millions of people worldwide, and many adults with diabetes remain untreated.

Rationale

- A systematic review of mainly sub-Saharan African studies *(45)* found that integrating HIV, diabetes and hypertension care into HIV services maintained high suppression of viral loads and good ART adherence, improved blood pressure control compared with non-integrated care and found some evidence for diabetes control. Important population health gains are expected and serious harm is very unlikely.
- Evidence from the United Republic of Tanzania suggests that people generally view integrated HIV, diabetes and hypertension care positively for reducing stigma, costs and service duplication and improving quality and early detection *(46)*.

7.6 Integrating HIV services and mental health care

Recommendation

Mental health care for depression, anxiety and alcohol use disorders should be integrated with HIV services.

(Strong recommendation, moderate-certainty evidence for depression and low-certainty evidence for anxiety and alcohol use disorder)

New

Background

People living with HIV commonly experience depression, anxiety, alcohol-use disorders and related stigma, which together worsen HIV outcomes. Integrated models that combine HIV, mental health and alcohol-use care are therefore needed to improve treatment success and well-being.

Rationale

- Integrating HIV care with mental health and alcohol-use services – at minimum by co-locating standardized HIV and mental health care in the same setting – can lead to a large reduction in moderate-to-severe depression and may reduce anxiety and alcohol use. Integration may also improve retention in HIV care and viral suppression *(47)*.
- WHO has identified a set of cost-effective mental health interventions with favourable returns on investment in terms of health and productivity (48). The evidence suggests that task-shared, integrated HIV and mental health care is likely to deliver benefits that outweigh implementation costs *(49, 50)*.

Implementation considerations: integrating HIV, diabetes, hypertension and mental health care

The following WHO resources are available to support the integration of HIV, diabetes, hypertension and mental health care:

- integration of noncommunicable diseases into HIV service packages *(51)*;
- integrating the prevention and control of noncommunicable diseases in HIV, TB and sexual and reproductive health programmes *(52)*;
- integration of mental health and HIV interventions: key considerations *(53)*; and
- WHO's Mental Health Gap Action Programme (mhGAP) operations manual *(54)*.

The decision to integrate diabetes and hypertension care into existing HIV programmes or to establish separate chronic disease clinics should be guided by factors such as the relative prevalence of conditions, contextual considerations, human resources and the capacity of current services. An international multi-stakeholder Delphi consensus study identified five essential items for effective integrated care for chronic conditions *(55)*:

- improved data collection and surveillance of diabetes and hypertension among people living with HIV;
- strengthened drug procurement systems;
- availability of equipment and access to diagnostics;
- health education; and
- enhanced continuity of care.

The chronic disease clinic model may offer a more effective and sustainable alternative to vertical programmes. It is important that all users of care services have access to care and treatment for diabetes, hypertension and mental health care regardless of their HIV status.

A differentiated service delivery approach should be applied to determine who provides services, what services are provided, where they are provided and when they are provided.

- **Who provides the services:** assessment of the need and opportunity for task-sharing to non-physician health-care workers should also be considered, including supervision and support across diseases. WHO hypertension guidance strongly supports task sharing to trained non-physician health-care workers and generally supports the integration of noncommunicable disease care into primary care *(56)*. mhGAP also strongly supports the provision of mental health care at primary care and through trained non-physician clinicians *(57)*.
- **What services are provided:** managing noncommunicable diseases and mental health conditions includes screening, detecting, treatment and providing access to palliative and rehabilitation services care for people in need. WHO clinical guidelines for managing diabetes, hypertension and mental health conditions are available *(56, 58–59)*.
- **Where are services provided:** integrated services may be decentralized to primary care and the community. Opportunities for out-of-facility services should be considered, especially for medication refill provision for both ART and managing noncommunicable diseases. Integrated service delivery should also be considered in prisons and other places of detention.
- **When are services provided:** services for people with diabetes and hypertension may also be differentiated based on whether the client is established on treatment. Clinical visits for all conditions may be aligned and reduced to every 3–6 months among established clients *(56)*, and multimonth dispensing should be the goal as for ART *(60)*.

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Annex

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2026 Consolidated HIV service delivery guidelines

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2 Updated recommendations on service delivery for the treatment and care of people living with HIV. Geneva: World Health Organization; 2021 (https://iris.who.int/handle/10665/341052). Licence: CC BY-NC-SA 3.0 IGO.

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2016 Consolidated guidelines on the use of antiretroviral drugs for treating and preventing HIV infection: recommendations for a public health approach[3]

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4 Consolidated guidelines on the use of antiretroviral drugs for treating and preventing HIV infection: recommendations for a public health approach, June 2013. Geneva: World Health Organization; 2013 (https://iris.who.int/handle/10665/85321). Licence: CC BY-NC-SA 3.0 IGO.

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Research gaps identified through the guideline development process

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- Client preferences for where to start ART
- Tools to support initiation outside the health facility
- Outcomes associated with less frequent clinical visits and/or drug refills (beyond six months) for various populations
- Tailored support to minimize disengagement and support re-engagement at different stages along the continuum of care
- Preferred methods of tracing and re-engagement for different population group (such as key populations), gender and age groups
- Integration approaches that improve uptake of sexual and reproductive health services, including contraception, and that incorporate cervical cancer screening and vaccination.
- Integration strategies across diverse health systems and social contexts, including provision of contraception alongside less frequent clinical and ART refill visits.
- Long-term data on the health outcomes of people living with HIV with noncommunicable diseases
- Cost–effectiveness of different models of integrated care
- Health promotion interventions to support lifestyle change and reduce noncommunicable disease risk among people living with HIV, including those facing stigma and access barriers
- Research to inform integration of hypertension and diabetes care within differentiated HIV service delivery models
- Values and preferences of people living with HIV and noncommunicable diseases regarding care delivery
- Research to inform feasible and effective training, supervision and scale-up of psychosocial support models, including those delivered by peer providers

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- Implementation research evaluation of different models of community-level support
- Cost of implementing community-level interventions in different settings
- Optimal ways to monitor adherence and identify clients in greatest need of adherence support
- Interventions to support adherence in populations at heightened risk of suboptimal adherence
- Combination adherence support interventions
- Feasibility and benefits of increasingly spaced clinical visits and medication pickup (greater than 6 months)
- Feasibility and safety of further task sharing to improve access to treatment, including initiation and maintenance of ART at community level
- Initiation of second-line ART by trained non-physician clinicians, midwives and nurses
- Cost–effectiveness of adolescent-friendly services
- Integration of SRH in ART services for adolescents
- Interventions to support safe disclosure

Guideline development methods

Retrieving, summarizing and presenting the evidence

All systematic reviews followed the PRISMA guidelines for reporting systematic reviews and meta-analyses. The GRADE (Grading of Recommendations Assessment, Development and Evaluation) approach was used to rate the certainty of evidence and determine the strength of recommendations. GRADE defines the certainty of evidence as the extent to which one can be confident that the reported estimates of effect (desirable or undesirable) available from the evidence are close to the actual effects of interest. After the evidence is collected and summarized, GRADE provides explicit criteria for rating the certainty of evidence that include study design, risk of bias, imprecision, inconsistency, indirectness and magnitude of effect. The strength of a recommendation reflects the degree to which the GDG is confident that the desirable effects (potential benefits) of the recommendation outweigh the undesirable effects (potential harm). Desirable effects may include beneficial health outcomes such as reduced morbidity and mortality, reduction of burden on the individual and/or health services and potential cost-savings. Undesirable effects include those affecting individuals, families, communities or health services as well as the implementation of services that may not be cost-effective within a particular context. Additional considerations include resource use and cost implications of implementing the recommendations and clinical outcomes (such as drug resistance and drug toxicity). The PRECIS-2 tool was used to assess the extent to which the conduct of a given study reflects an explanatory (ideal situation) or a more pragmatic (usual care) context. PRECIS-2 has nine domains – eligibility criteria, recruitment, setting, organization, flexibility (delivery), flexibility (adherence), follow-up, primary outcome and primary analysis – scored from 1 (very explanatory) to 5 (very pragmatic). The level of integration for interventions identified through systematic review was classified using the Standard Framework for Levels of Integrated Healthcare.

Feasibility and acceptability

The pragmatism of studies provided indirect evidence on feasibility from the systematic reviews, obtained using PRECIS-2. Other sources of evidence for feasibility included the UNAIDS database on country adoption of relevant WHO recommendations. Surveys were conducted among people living with HIV to assess the feasibility and acceptability of strategies for delivering HIV services. Dissemination of this survey was supported by the Global Network of People Living with HIV (GNP+).

Resource use and cost-effectiveness

The systematic reviews captured published evidence on resource use including costing, cost-effectiveness and affordability data. Additional literature reviews were conducted for indirect evidence that could enrich this information. The GDG and External Review Group included representatives from national programmes who also provided perspectives on resource implications in their countries.

Ethical considerations

Before the GDG meetings, the proposed areas of intervention were reviewed by a WHO staff member with expertise in global health and ethics, and key issues with respect to equity were outlined to the group for consideration when formulating recommendations.

Peer review

The draft guidelines were circulated for review to members of the GDG and External Review Group. The WHO Guideline Steering Group reviewed the comments and incorporated them into the final document with due consideration for any conflicts of interest relating to External Review Group members.

Declarations of interest

All external contributors to the guidelines, including members of the GDG and External Review Group, completed a WHO declaration of interests form in accordance with WHO policy for experts. A brief biography of each GDG member was published on the WHO HIV website for a period of 14 days before the first meeting of the group with a description of the objectives of the meeting. No public comments or objections were received. The responsible technical officer reviewed the declaration of interests forms as well as the results of the web-based search for each member of the GDG. The results were shared with the WHO Guideline Steering Group, which reviewed the results, and a management plan was agreed and recorded for each individual. At the start of the guideline development meeting, all conflicts of interest identified and the management plan for any conflicts of interest were shared with the meeting participants and members were asked to vocalize any additional conflicts or undeclared conflicts.
In accordance with the revised WHO policy for experts, a web-based search of GDG members was conducted to identify any potential competing interest. The WHO Guideline Steering Group recorded and reviewed the results of the web-based search to identify any potential competing interest. There were no conflicts of interest that warranted exclusion from the discussion of specific recommendations. Similarly, for the External Review Group the responsible technical officers reviewed all declaration of interest forms in accordance with WHO guideline development policy. Any conflicts of interest identified were considered when interpreting comments from External Review Group members during the external review process.

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